

Advanced Econometrics

Sample selection

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Sample selection

- Sample selection or self-selection problem
 - People in a sample may be different than the rest of the population (e.g. expected salaries based on employed only, PD modeling using KGB only etc.)
 - in such a case casual effect (for the whole population) cannot be estimated unless selection problem is solved
 - otherwise, we will never know which part of the observable outcome is related to the causal relationship and which part is due to the fact that different people were included in a sample and different excluded.

Solving sample selection issue

- Solving sample selection problems requires:
 - **Random experiment**, the situation where the participants do not have control over their status but the researcher does. Randomization is often the best possible method as it is easy to analyze and understand. However, this method is seldom feasible for practical and ethical reasons.
 - **Instruments (exclusion restrictions)** - variables, observable to the researcher, and which determine the treatment status but not the outcome.
 - Information about the **functional form of the selection and outcome processes**, such as the distribution of the disturbance terms. The original Heckman's solution belongs to this group. However, the functional form assumptions are usually hard to justify.
- Randomization or pseudo-randomization (natural experiments) have become state of the art for estimating causal effects.
- However, methods relying on distributional assumptions are still widely used.
 - Method is simple and widely available
 - However!, reasonable instruments and parametric assumptions hard to justified – real vs artificial (based on assumptions) effect.
 - Design of a study is not in hand of a researcher

Heckman sample selection model

- Sometimes HSS and its generalizations are also called *generalized Tobit models*
- Standard Heckman sample selection model (Tobit-2 model)
 - Latent (unobserved) structural process:

$$\begin{aligned}y_i^{S*} &= \beta^S x_i^S + \varepsilon_i^S \\ y_i^{O*} &= \beta^O x_i^O + \varepsilon_i^O\end{aligned}$$

- y_i^{S*} the realization of the latent value of the selection tendency
- y_i^{O*} is the latent outcome.
- x_i^S and x_i^O independent variables (may be or not equal)
- We observed:

$$\begin{aligned}y_i^S &= \begin{cases} 0 & y_i^{S*} < 0 \\ 1 & y_i^{S*} \geq 0 \end{cases} \\ y_i^O &= \begin{cases} 0 & y_i^S = 0 \\ y_i^{O*} & y_i^S = 1 \end{cases}\end{aligned}$$

Properties of a model

- If $y_i^{S*} \geq 0$ then The observed dependence between y_i^S and y_i^O can be written as:

$$E(y^O | x^O = x_i^O, x^S = x_i^S, y^S = 1) = \beta^O x_i^O + E(\varepsilon^O | \varepsilon^S \geq -\beta^S x_i^S)$$

- Estimating the model by OLS gives in general biased results ($E(\varepsilon^O | \varepsilon^S \geq -\beta^S x_i^S) \neq 0$)
- unless ε^O and ε^S are mean independent ($\rho = 0$)

- Assuming $\begin{pmatrix} \varepsilon^O \\ \varepsilon^S \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right)$ a simple strategy may be employed:

- Find the expectations $E(\varepsilon^O | \varepsilon^S \geq -\beta^S x_i^S)$ - control function, by estimating probit
- And insert the expectation into outcome equation:

$$y^O = \beta^O x_i^O + E(\varepsilon^O | \varepsilon^S \geq -\beta^S x_i^S) + \eta_i \equiv \beta^O x_i^O + \rho\sigma\lambda(\beta^S x_i^S) + \eta_i$$

- where $\lambda(\cdot) = \phi(\cdot)/\Phi(\cdot)$ - inverted Mill's ratio (pdf/cdf of normal distribution)
- $\rho\sigma$ – estimated by OLS
- Essentially, selection problem is omitted variable problem in this framework
- OLS formula for the coefficient variance-covariance matrix is not appropriate (other consistent method is proposed)

Two methods of estimation

- Methods:
 - 2-step approach
 - Probit & OLS – step-by-step
 - Pros:
 - the two-step solution allows certain generalizations more easily than ML,
 - and is more robust in certain circumstances
 - Cons:
 - Suboptimal solution
 - 1-step MLE estimation
 - Note: The model is **well identified** if the **exclusion restriction is fulfilled**, i.e. if selection regression includes a component with a substantial explanatory power but which is not present in outcome regression (otherwise some semiparametric methods of estimations should be employed)

Marginal effects after Heckman model

- 3 type of marginal effects may be analyzed:
 - $\frac{\delta E(y_{oi}|x_{oi})}{\delta x_{oi}} = \beta_i$ - increase of expected value of y for randomly selected observation for given x
 - $\frac{\delta P(y_{si} = 1|x_{si})}{\delta x_{si}}$ – increase of probability of being selected
 - $\frac{\delta E(y_{oi}^+|x_{oi}, x_{si}, y_{si} = 1)}{\delta x_{oi}}$ – increase of expected value of y assuming y=0 for not selected, and y for selected for given x
- Example:
 - We are analyzing salary at a market:
 - $E(y_{oi}|x_{oi})$ – is an expected value of salary if everybody would work for a given x (we may say it is an offering salary for a given x)
 - $P(y_{si} = 1|x_{si})$ – is an probability that somebody works for a given x
 - $E(y_{oi}^+|x_{oi}, x_{si}, y_{si} = 1)$ – is an expected value of salary for a given x (assuming that some part of observations does not work and have salary=0)